

CSES – Interactive Problems Reference

All tasks: 1 s, 512 MB. Flush after every output line; respect the query budget and terminate immediately when finished.

3112 – Hidden Integer

A hidden integer x is fixed. Determine x by asking threshold questions of the form “is $y < x$?”. **Start:** no initial input; begin querying immediately

Ask: print ? y , $1 \leq y \leq 10^9$

Reply: YES iff $y < x$, otherwise NO

Finish: print ! x and terminate

Constraints: $1 \leq x \leq 10^9$; at most 30 ? queries.

3139 – Hidden Permutation

A hidden permutation $a_1, \dots, a_n$ of $1, \dots, n$ is fixed. Recover every value by comparing entries at chosen indices. **Start:** read n

Ask: print ? i j to ask whether $a_i < a_j$

Reply: YES iff $a_i < a_j$, otherwise NO

Finish: print ! $a_1 \dots a_n$ and terminate

Constraints: $n \leq 1000$; at most 10^4 comparison queries.

3305 – K-th Highest Score

There are n distinct Finnish scores and n distinct Swedish scores. Within each country, query scores by descending rank; find the k -th highest score among all $2n$ coders. **Start:** read n, k

Ask: print F i or S i for that country’s i -th highest score

Reply: the requested integer score

Finish: print ! s , where s is the overall k -th highest score, then terminate

Constraints: $n \leq 10^5$, $1 \leq k \leq 2n$; all $2n$ scores distinct, in $[1, 10^9]$; at most 100 score queries total.

3228 – Permuted Binary Strings

A hidden permutation $a_1, \dots, a_n$ is fixed. For any chosen binary string $b_1 \dots b_n$, the grader returns the string permuted as $b_{a_1} \dots b_{a_n}$. Recover the permutation. **Start:** read n

Ask: print ? b , where $b \in \{0, 1\}^n$

Reply: the length- n string $b_{a_1} \dots b_{a_n}$

Finish: print ! $a_1 \dots a_n$ and terminate

Constraints: $n \leq 1000$; at most 10 binary-string queries.

3273 – Colored Chairs

n chairs form a cycle; every chair is red or blue, and n is odd. Find an index i such that adjacent chairs i and $i + 1$ have the same color, with $n + 1 \equiv 1$. **Start:** read n

Ask: print ? i to query chair i

Reply: R or B

Finish: print ! i asserting chairs i and $i + 1$ have equal color, then terminate

Constraints: $3 \leq n \leq 2 \cdot 10^5$, n odd; at most 20 color queries.

3140 – Inversion Sorting

A hidden permutation of $1, \dots, n$ must be sorted. Each move reverses a chosen subarray; afterward the grader reports the permutation’s new inversion count. You win as soon as it becomes 0. **Start:** read n

Ask: print i j to reverse positions $i \dots j$

Reply: the inversion count after that reversal

Finish: when the returned count is 0, terminate immediately; there is no separate ! command

Constraints: $n \leq 1000$; at most $4n$ reversal operations.
